Ln 1, Col 1

TStates : 179 | MCycles : 53 | Time of Execution : 179 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
00	5A	27	00	00	00	00	00	47FF	4100

```
LXI SP 47FFh
LDA 4500h
MOV B A
LDA 4501h
MOV C A
MOV A B
ADD C
STA 4502h
MVI A 00h
ACI 00h
STA 4503h
LDA 4500h
MOV B A
LDA 4501h
MOV C A
MOV A B
SUB C
STA 4504h
MVI A 00h
ACI 00h
STA 4505h
HLT
```

Starting Address : 4100

Comments :

User Data Grid

Address	Data
4500	5A
4501	27
4502	81
4503	00
4504	33
4505	00
4510	34
4511	12
4512	21
4513	43
4516	00
4517	00
4518	00
4519	00
451A	00

Hex Code Grid

Address	Data
4100	31
4101	FF
4102	47
4103	3A
4104	00
4105	45
4106	47
4107	3A
4108	01
4109	45
410A	4F
410B	78
410C	81
410D	32
410E	02
410F	45
4110	3E
4111	00

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

Code Pad

Ln 27, Col 1

TStates : 231 | MCycles : 69 | Time of Execution : 231 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
01	12	43	12	34	55	55	00	0000	4100

```
LHLD 4510h
XCHG
LHLD 4512h
DAD D
SHLD 4514h
MVI A 00h
ACI 00h
STA 4516h
LDA 4510h
MOV B A
LDA 4512h
MOV C A
MOV A B
SUB C
STA 4518h
LDA 4511h
MOV B A
LDA 4513h
MOV C A
MOV A B
SBB C
STA 4519h
MVI A 00h
ACI 00h
STA 451Ah
HLT
```

Starting Address : 4100

Comments :

User Data Grid

Address	Data
4510	34
4511	12
4512	21
4513	43
4514	55
4515	55
4516	00
4518	13
4519	CF
451A	01

Hex Code Grid

Address	Data
4100	2A
4101	10
4102	45
4103	EB
4104	2A
4105	12
4106	45
4107	19
4108	22
4109	14
410A	45
410B	3E
410C	00
410D	CE
410E	00
410F	32
4110	16
4111	45

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

MOVe -- copy from source to destination

Ln 17, Col 9

TStates : 195 | MCycles : 57 | Time of Execution : 195 µs
Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
00	E8	9C	00	00	4C	00	00	47FF	8000

```
LXI SP 47FFh
LDA 4500h
MOV B A
LDA 4501h
ADD B
PUSH PSW
STA 4502h
POP H
MOV A L
STA 4503h
LDA 4500h
MOV B A
LDA 4501h
MOV C A
MOV A B
SUB C
PUSH PSW
STA 4504h
POP H
MOV A L
STA 4505h
HLT
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4500	E8
4501	9C
4502	84
4503	95
4504	4C
4505	00

Hex Code Grid

Address	Data
8000	31
8001	FF
8002	47
8003	3A
8004	00
8005	45
8006	47
8007	3A
8008	01
8009	45
800A	80
800B	F5
800C	32
800D	02
800E	45
800F	E1
8010	7D
8011	32

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

RETurn unconditionally

Ln 22, Col 12

TStates : 411 | MCycles : 119 | Time of Execution : 411 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
03	06	03	00	14	00	78	00	0000	8000

```
LDA 4520h
MOV E A
MVI D 00h
LDA 4521h
MOV C A
LXI H 0000h
MOV A C
ORA A
JZ MULT_DONE
MULT_LOOP: DAD D
DCR C
JNZ MULT_LOOP
MULT_DONE: SHLD 4522h
LDA 4525h
MOV B A
MOV A B
ORA A
JZ DIV_ZERO
LDA 4524h
MVI C 00h
DIV_LOOP: CMP B
JC DIV_DONE
SUB B
INR C
JMP DIV_LOOP
DIV_DONE: STA 4527h
MOV A C
STA 4526h
HLT
DIV_ZERO: MVI A 0Fh
STA 4540h
HLT
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4520	14
4521	06
4524	14
4525	06
4522	78
4523	00
4526	03
4527	02

Hex Code Grid

Address	Data
8000	3A
8001	20
8002	45
8003	5F
8004	16
8005	00
8006	3A
8007	21
8008	45
8009	4F
800A	21
800B	00
800C	00
800D	79
800E	B7
800F	CA
8010	17
8011	80

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmetic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

LoaD H and L register direct

Ln 29, Col 11

TStates : 8251 | MCycles : 2148 | Time of Execution : 8251 µs
Flags : S Z AC P CY

General Regs.:	A	B	C	D	E	H	L	M	SP	PC
	FF	00	85	00	25	00	85	00	47FF	8000

```
LXI SP 47FFh
LHLD 4530h
XCHG
LHLD 4532h
MOV B H
MOV C L
LXI H 0000h
SHLD 4536h
MOV A B
ORA C
JZ STORE_PRODUCT
MUL_LOOP: DAD D
JNC NO_CARRY
PUSH H
LHLD 4536h
INX H
SHLD 4536h
POP H
NO_CARRY: DCX B
MOV A B
ORA C
JNZ MUL_LOOP
STORE_PRODUCT: SHLD 4534h
LHLD 453Ah
MOV A H
ORA L
JZ DIV_ZERO
XCHG
LHLD 4538h
LXI B 0000h
DIV_LOOP: MOV A L
SUB E
MOV L A
MOV A H
SBB D
MOV H A
JC DIV_DONE
INX B
JMP DIV_LOOP
DIV_DONE: DAD D
SHLD 453Eh
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4530	57
4531	13
4532	25
4533	00
4538	57
4539	13
453A	25
453B	00
4534	93
4535	CB
4536	02
4537	00
453C	85
453D	00
453E	1E
453F	00

Hex Code Grid

Address	Data
8000	31
8001	FF
8002	47
8003	2A
8004	30
8005	45
8006	EB
8007	2A
8008	32
8009	45
800A	44
800B	4D
800C	21
800D	00
800E	00
800F	22
8010	36
8011	45

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmetic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

Code Pad

Ln 19, Col 8

TStates : 66 | MCycles : 20 | Time of Execution : 66 µs

Flags : S Z AC P CY

General Regs.: A 0F | B C 00 00 | D E 00 00 | H L 00 00 | M 00 | SP 0000 | PC 8000

```
LDA 453Ah
MOV B A
LDA 453Bh
ORA B
JNZ VALID
MVI A 0Fh
STA 4540h
HLT
VALID: LXI SP 47FFh
LHLD 453Ah
MOV A H
ORA L
JZ ERROR
XCHG
LHLD 4538h
LXI B 0000h
DIVLOOP: MOV A L
SUB E
MOV L A
MOV A H
SBB D
MOV A H
SBB D
MOV H A
JC DIVDONE
INX B
JC DIVLOOP
DIVDONE: DAD D
SHLD 453Ch
MVI A 00h
STA 4540h
HLT
ERROR: MVI A 0Fh
```

Starting Address : 8000

Comments :

User Data Grid

Address	Data
4538	34
4539	12
453A	00
453B	00
4540	0F

Hex Code Grid

Address	Data
8000	3A
8001	3A
8002	45
8003	47
8004	3A
8005	3B
8006	45
8007	B0
8008	C2
8009	11
800A	80
800B	3E
800C	0F
800D	32
800E	40
800F	45
8010	76
8011	31

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

LoaD Accumulator indirect

Ln 14, Col 4

TStates : 252 | MCycles : 74 | Time of Execution : 252 µs

Flags : S Z AC P CY

General Regs.:

A	B	C	D	E	H	L	M	SP	PC
07	00	00	00	FB	06	DD	00	0000	4300

```
LDA 4520h
MOV E A
MVI D 00h
LDA 4521h
MOV C A
LXI H 0000h
MOV A C
ORA A
JZ MULTDONE
MULTLOOP: DAD D
DCR C
JNZ MULTLOOP
MULTDONE: SHLD 4522h
HLT
```

Starting Address : 4300

Comments :

User Data Grid

Address	Data
4520	FB
4521	07
4522	DD
4523	06

Hex Code Grid

Address	Data
4300	3A
4301	20
4302	45
4303	5F
4304	16
4305	00
4306	3A
4307	21
4308	45
4309	4F
430A	21
430B	00
430C	00
430D	79
430E	B7
430F	CA
4310	17
4311	43

1.Data Transfer Group :

MOV	MVI	LDA	STA
LHLD	SHLD	LXI	XCHG
LDAX	STAX	IN	OUT

2.Arithmatic Group :

ADD	ADI	ADC	ACI	DAA
SUB	SUI	SBB	SBI	DAD
INR	DCR	INX	DCX	

3.Logical Group :

ANA	ANI	XRA	XRI	CMA
ORA	ORI	CMP	CPI	CMC
RLC	RRC	RAL	RAR	STC

4.Branch and Machine Control Group

JMP	J_	CALL	C_	
RET	R_	EI	DI	PUSH
PCHL	RST	RIM	SIM	POP
XTHL	SPHL	NOP	HLT	

Auto Insert | Insert | Cancel

Instruction Information :

Code Pad